INCH-POUND
MIL-DTL-24643/4D
22 August 2002
SUPERSEDING
MIL-C-24643/4C
14 March 1994

DETAIL SPECIFICATION SHEET

CORD, ELECTRICAL, 600 VOLTS, TYPE LSMCOS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First -	Copper conductor, uncoated. (see table I for size)
Second -	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third -	Ethylene propylene rubber or cross-linked polyethylene insulation. (see table I for wall thickness) - Sizes 2, 4 and 7 shall have special identification code applied by method 3.- Size 5 shall have one shielded pair (colors black and white) and 3 singles (colors red, green and blue)applied by method 3.- Size 6 shall have two shielded pairs (colors black and white for first pair and blue and red of second pair) and two singles (colors yellow and green) applied by method 3.
Fourth -	Braided shield (for all pairs) of AWG No. 34 or No. 36 uncoated copper. Braid angle 30 to 35 degrees; minimum coverage of 88 percent.
Fifth -	The required number of singles and shielded pairs with a left-hand lay and with a minimum amount of fillers to make a round assembly. The length of lay shall be determined to permit the completed cord to withstand the twisting and bending endurance test, and for sizes 5 and 6, the mutual inductance tests. The size 7 shall have a center of glass. A single conductor shall not be used as a central core for the cord.

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Construction (non-watertight) (continued)

Sixth - In size 5 and 6 only, a binder tape applied helically with overlap. In sizes 2, 4 and 7: first a binder; second, a braided shield of AWG No. 34 or AWG No. 36 uncoated strands, having a braid angle of 30 degrees minimum, and providing a coverage of 88 percent minimum; third a separator tape.

Seventh - Cross-linked polyolefin jacket (black). (see table I for diameters)

TABLE I. Details

Military part no. M24643/4	Type and Size	Conductors		Insulation thickness minimum average (inch)	Overall diameter		Voltage withstand		Insulation resistance for 1000 feet minimum (megohms)	Conductor resistance per 1000 feet maximum (ohms)
		No. prs.	AWG		min (inch)	max (inch)	Conductor to conductor minimum (volts)	Shield to ground minimum (volts)		
-01UO	LSMCOS-2	2	18 (class K)	0.023	0.440	0.460	2000	1500	200	6.96
-02UO	LSMCOS-4	4	18 (class K)	0.023	0.490	0.510	2000	1500	200	6.96
-03UN	LSMCOS-5	5	20 (class K)	0.013	0.375	0.390	1000	500	100	11.28
-04UN	LSMCOS-6	6	20 (class K)	0.013	0.460	0.480	1000	1000	100	11.28
-05UO	LSMCOS-7	7	18 (class K)	0.023	0.575	0.595	2000	1500	200	6.96

EXAMINATION AND TESTS:

Basic electrical: Requirements:

Conductor resistance - ohms/1000 feet at 25°C, maximum .....(see table I)
Voltage withstand - volts, root mean square, minimum
Conductor to conductor .....(see table I)
Shield to ground.....(see table I)
Insulation resistance – conductor to shield, megohms -1000 feet, minimum.....(see table I)
Conductor and shield continuity ..... No failure

NOTE: Shield continuity applicable to LSMCOS-2, LSMCOS-4, and LSMCOS-7 only.

Group A:

Visual and dimensional..... No failure

Group B:

Cross-linked proof test (percent, maximum)
Insulation 50
Jacket (When tested at 200°C).....50
Tear - pounds per inch thickness, minimum (ASTM D 470) 35

Physicals (unaged)

Insulation
Ethylene propylene rubber
Tensile strength - lb/in^2, minimum 700
Elongation - percent, minimum. 250

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Cross-linked polyethylene
Tensile strength - lb/in^2,minimum 1800
Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Twisting and bending endurance – (see note 2)
At 75°C, cycles, minimum.....6000
At minus 30 ± 2°C, cycles, minimum.....2000

NOTE: 2. Roller diameters for bending endurance test shall be as follows:

Cable type and size	Roller Diameter (inch minimum)
LSMCOS-2	1.4
LSMCOS-4	1.5
LSMCOS-5	1.2
LSMCOS-6	1.4
LSMCOS-7	1.8

Cold bending, cord at minus 40 ± 2°C..... No failure

Group C:
Mutual inductance (LSMCOS-5 and LSMCOS-6 only)
Microhenries, maximum 0.0135
Physicals (aged)
Insulation
Ethylene propylene
Air oven
Tensile strength - percent of unaged, minimum 75
Elongation - percent of unaged minimum 75
Cross-linked polyethylene
Air oven
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Shrinkage ..... No failure
Permanence of printing (jacket) - cycles, minimum.....125
Heat distortion - percent of unaged, maximum 30

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Group D:

Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements:
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH: 2000 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-003)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

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